

3(a)	<u>sum/total</u> momentum (of system of bodies) is constant or <u>sum/total</u> momentum before = <u>sum/total</u> momentum after	M1
	for an isolated system/no (resultant) <u>external</u> force	A1
3(b)(i)	$p = mv$	C1
	$(4.0 \times 6.0 \times \sin \theta) - (12 \times 3.5 \times \sin 30^\circ) = 0$ or $(m_A v_A \times \sin \theta) - (m_B v_B \times \sin 30^\circ) = 0$	M1
	$\theta = 61^\circ$	A1

Question	Answer	Marks
3(b)(ii)	shows the horizontal <u>momentum</u> component of ball A or of ball B as $(4.0 \times 6.0 \times \cos \theta)$ or $(12 \times 3.5 \times \cos 30^\circ)$	C1
	$(4.0 \times 6.0 \times \cos 61^\circ) + (12 \times 3.5 \times \cos 30^\circ) = 4.0v$ so $v = 12 \text{ (ms}^{-1}\text{)}$	A1
3(b)(iii)	initial $E_K (= \frac{1}{2} \times 4.0 \times 12^2) = 290 \text{ (288) (J)}$	M1
	final $E_K (= \frac{1}{2} \times 4.0 \times 6.0^2 + \frac{1}{2} \times 12 \times 3.5^2) = 150 \text{ (145.5) (J)}$	M1
	(initial $E_K >$ final E_K) so inelastic [both M1 marks required to award this mark]	A1

Question	Answer	Marks
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